

# Engineering Physics

Aaryan Khedekar

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## 1 Quantum Theory

### 1.1 Introduction

1. This theory deals with the particles of size  $10^{-10}\text{m}$ .
2. We know that  $E = h\nu$  where  $h$  is the planck's constant.
3. Also the dimension of  $h = [ML^2T^{-2}]$  which is equal to that of the angular momentum.
4. Hence, if the angular momentum of any material is of order of  $h$  (i.e.  $10^{-34}$ ), then quantum mechanics is applicable to it.

### 1.2 Black Body

1. A black body is a hypothetical perfect absorber and radiator of energy, with no reflecting power.
2. Any naturally available object is not a black body, but we can make artificial black body to some extent. For example, Ferry's black body.
3. On heating a black body, it will emit all frequencies of the absorbed energy, in all directions.
4. The black body phenomenon can be explained by quantum mechanics, but prior to it, two classical theories were discovered, but failed due to the same reason, which will be discussed later.

#### 1.2.1 Rayleigh - Jean Law:

$$E_{\lambda}d\lambda = \frac{8\pi k_B T}{\lambda^4}d\lambda$$

### 1.2.2 Wein's Formula:

$$E_{\lambda}d\lambda = \frac{8\pi hc}{\lambda^5} e^{-\frac{hc}{\lambda k_B T}} d\lambda$$

Both of these fail to give a result when the wavelength tends to infinity, the energy becomes zero, which is not possible.

$$\lambda \rightarrow \infty \implies E \rightarrow 0$$

### 1.3 Planck's Radiation Law

1. A black body radiator contain simple harmonic oscialltor which may vibrate at a suitable frequency.
2. Energy of a blackbody radiator with frequency  $\nu$  can't be continuous; it is limited to discrete volume of a box called "quanta".
3. The particle represtenting quanta of light is called as "photon". The energy of a photon is equal to  $\alpha$  to  $\gamma$  radiation.  $E = h\nu$
4. An oscialltor frequency will absorb energy and will be of discrete energy levels.  $h\nu, 2h\nu, 3h\nu, \dots$
5. The energy density (energy per unit volume) will be in the range of  $\lambda$  to  $\lambda + d\lambda$ .
- 6.

$$E_{\lambda}d\lambda = \frac{8\pi hc}{\lambda^5} \cdot \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1} d\lambda$$

- (a) **At low temperature**,  $\lambda k_B T$  will be very small  $\implies e^{\frac{hc}{\lambda k_B T}} \gg 1$ .

Hence,

$$E_{\lambda}d\lambda = \frac{8\pi hc}{\lambda^5} e^{-\frac{hc}{\lambda k_B T}} d\lambda$$

This is the **Wien's Law**.

- (b) **At very high temperature**,

$$e^{\frac{hc}{\lambda k_B T}} = 1 + \frac{hc}{\lambda k_B T} + \frac{1}{2!} \left( \frac{hc}{\lambda k_B T} \right)^2 + \dots \approx 1 + \frac{hc}{\lambda k_B T}$$

Hence,

$$E_{\lambda}d\lambda = \frac{8\pi k_B T}{\lambda^4} d\lambda$$

This is the **Rayleigh-Jean's Law**.

7. **Planck's Quanta Hypothesis** leads to the conclusion that the radiation is not continuous, but discrete

## 1.4 Properties of Photons

1. **Energy of Photon:**  $E = h\nu$
2. **Mass of Photon:** We know  $E = h\nu$  and according to Einstein's mass - energy relation,  $E = mc^2$ , we get  $m = \frac{h\nu}{c^2}$
3. **Momentum of Photon:**  $p = mv = (\frac{h\nu}{c^2})(c) \implies p = \frac{h\nu}{c}$
4. **Charge of Photon:** A photon is neutral.
5. **Wave-Particle Duality:** One photon can be considered as a particle, but a stream of photons is considered as a wave.

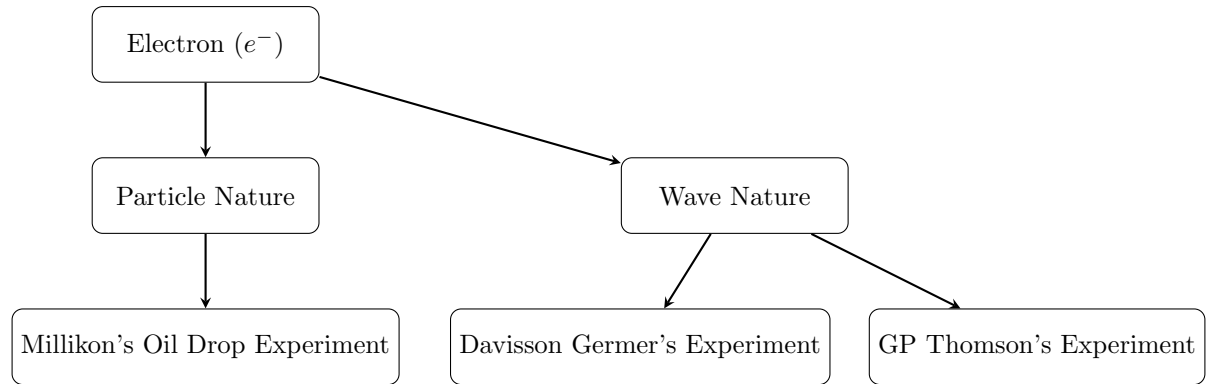
## 1.5 Photoelectric Effect

1. It is phenomenon in which electrons gets out of the metal surface when light is incident on it.
2. The metal is called a "photosensitive material" and the electrons are called as "photoelectrons".
3. Electron emission increases with increase of light intensity. For each substance, there is a minimum frequency  $\nu_0$  for photoelectrons to emit. This is called as the **threshold frequency**.
4. In order for an electron to escape from the metal surface, the incident light must have enough energy, which is called as the **work function** ( $\phi$ ).
  - (a) **When**  $E = \phi(\nu = \nu_0)$ , electrons only come up to the surface of the metal, but doesn't get emitted. ( $KE = 0$ ).
  - (b) **When**  $E < \phi(\nu < \nu_0)$ , electrons doesn't get emitted, they will remain inside the metal surface. ( $KE = 0$ ).
  - (c) **When**  $E > \phi(\nu > \nu_0)$ , electrons get emitted out of the metal surface. ( $KE = \frac{1}{2}mv_{max}^2$ ).
5. **Stopping Potential:**  $eV_s = KE = h(\nu - \nu_0) \implies V_s = \frac{h}{e}(\nu - \nu_0)$

## 1.6 Compton Effect

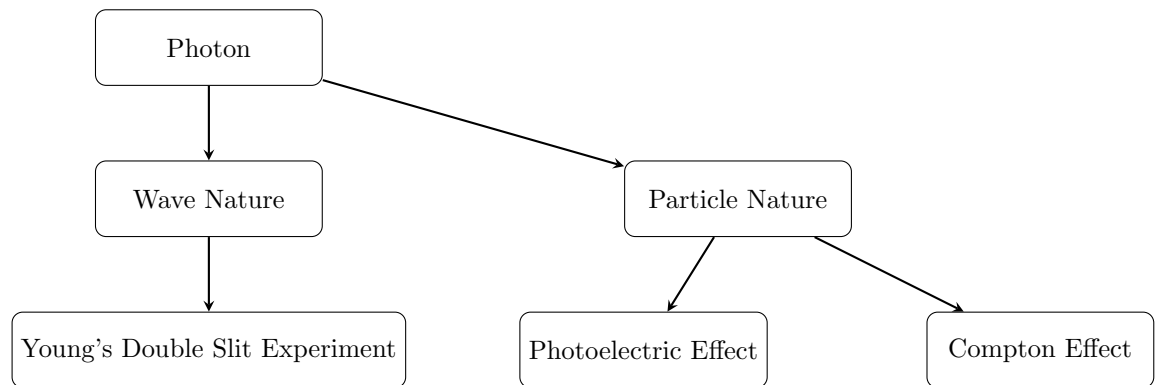
### 1.6.1 Electrons:

An electron is a particle, which behaves as a wave under certain situations.



### 1.6.2 Photons:

An photon is a wave, which behaves as a particle under certain situations.



### 1.6.3 Properties of Photons as Particle:

1. Photons have mass:  $E = mc^2 \implies m = \frac{E}{c^2} = \frac{h\nu}{c^2}$
2. Photons have momentum:  $p = mc = \frac{E}{c} \implies p = \frac{h\nu}{c}$
3. Photons have no charge, but when they fall on metal surface, they emit electrons.
4. Spin of a Photon = 1

#### 1.6.4 Derivation of Compton Effect:

Let a photon be incident on an electron on rest. On collision, the electron will be recoiled (assuming the photon has less energy than excitation energy of electron), and the photon will get scattered elsewhere. Let the photon be scattered at an angle of  $\theta$  (scattering angle) and the electron gets recoiled at an angle of  $\phi$  (recoil angle)

**By Conservation of Energy,**

$$h\nu + m_0c^2 = h\nu' + mc^2$$

$$(h\nu - h\nu') + m_0c^2 = mc^2$$

Squaring both sides,

$$(h\nu - h\nu')^2 + m_0^2c^4 + 2m_0c^2h(\nu - \nu') = m^2c^4$$

$$h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu' + m_0^2c^4 + 2m_0c^2h(\nu - \nu') = m^2c^4$$

Dividing by  $c^2$  on both sides,

$$\frac{h^2}{c^2}(\nu^2 + \nu'^2 - 2\nu\nu') + m_0^2c^2 + 2m_0h(\nu - \nu') = m^2c^2$$

But we know  $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} \implies m^2c^2 - m_0^2c^2 = m^2v^2$

$$\frac{h^2}{c^2}(\nu^2 + \nu'^2 - 2\nu\nu') + 2m_0h(\nu - \nu') = m^2v^2 - (1)$$

**By Conservation of Momentum,**

$$\frac{h\nu}{c} + 0 = \frac{h\nu'}{c}\cos\theta + m\nu\cos\phi - (2)$$

$$\frac{h\nu'}{c}\sin\theta = m\nu\sin\phi - (3)$$

Squaring and adding (2) and (3),

$$\frac{h^2\nu^2}{c^2} + \frac{h^2\nu'^2}{c^2}\cos^2\theta + \frac{h^2\nu'^2}{c^2}\sin^2\theta - 2\frac{h^2}{c^2}\nu\nu'\cos\theta = m^2v^2\cos^2\phi + m^2v^2\sin^2\phi$$

$$\frac{h^2}{c^2}(\nu^2 + \nu'^2 - 2\nu\nu'\cos\theta) = m^2v^2 - (4)$$

Subtracting equation (1) from (4),

$$\frac{h^2}{c^2}(2\nu\nu' - 2\nu\nu'\cos\theta) - 2m_0h(\nu - \nu') = 0$$

$$\frac{h}{c^2}\nu\nu'(1 - \cos\theta) = m_0(\nu - \nu')$$

$$\Delta\nu = \nu - \nu' = \frac{h\nu\nu'}{m_0c^2}(1 - \cos\theta)$$

**This is the Compton Shift in terms of Frequency.**

Now,

$$\nu = \frac{c}{\lambda} \implies \frac{c}{\lambda} - \frac{c}{\lambda'} = \frac{hc^2}{m_0c^2\lambda\lambda'}(1 - \cos\theta)$$

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0c}(1 - \cos\theta)$$

**This is the Compton Shift in terms of Wavelength.**

From (2):  $p - p'\cos\theta = p_e\cos\phi$

from (3):  $p'\sin\theta = p_e\sin\phi$

Dividing (3) by (2),

$$\tan\phi = \frac{p'\sin\theta}{p - p'\cos\theta} \implies \phi = \tan^{-1}\left(\frac{p'\sin\theta}{p - p'\cos\theta}\right)$$

Multiplying and Dividing by c,

$$\phi = \tan^{-1}\left(\frac{p'c\sin\theta}{pc - p'ccos\theta}\right) \implies \phi = \tan^{-1}\left(\frac{E'\sin\theta}{E - E'\cos\theta}\right)$$

## 1.7 Matter Waves

1. e/m ratio strongly suggests that electrons have particle nature.
2. Light shows wave and particle nature.
3. Elemental particles like protons, neutrons, electrons show wave-like nature.
4. According to de-broglie's hypothesis, any moving particle associated with waves is called a "matter wave" or "de-broglie wave".

### 1.7.1 de-Broglie's Wavelength:

Consider any moving particle with velocity 'v'. Then according to de-Broglie's hypothesis,

$$\lambda \propto \frac{1}{p} \implies \lambda = \frac{h}{p} = \frac{h}{mv}$$

**Derivation:**

According to Planck's Theory,  $E = h\nu$

According to Einstein's Theory:  $E = mc^2$

$$h\nu = mc^2 \implies \frac{hc}{\lambda} = mc^2 \implies \lambda = \frac{h}{mc} = \frac{h}{p}$$

### 1.7.2 de-Broglie Wavelength of an accelerated charged particle:

We know  $KE = eV$  and  $KE = \frac{1}{2}mv^2$

$$eV = \frac{1}{2}mv^2 \implies v = \sqrt{\frac{2eV}{m}}$$
$$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{\frac{2eV}{m}}} \implies \boxed{\lambda = \frac{h}{\sqrt{2meV}}}$$

### 1.7.3 de-Broglie Wavelength in terms of Kinetic Energy:

$$KE = \frac{p^2}{2m} \implies p = \sqrt{2mKE} \implies \boxed{\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mKE}}}$$

### 1.7.4 de-Broglie Wavelength of a particle in thermal equilibrium:

We know  $KE = \frac{3}{2}kT$

$$\lambda = \frac{h}{\sqrt{2mKE}} = \frac{h}{\sqrt{2m(\frac{3}{2}kT)}} \implies \boxed{\lambda = \frac{h}{\sqrt{3mkT}}}$$

## 1.8 Davisson-Germer Experiment

1. This experiment explains the wave nature of electrons by the phenomenon of diffraction.
2. Consider an electron beam incident to a Nickel Crystal, which gets scattered at an angle of  $\phi$  to a scale with a movable detector.
3. The intensity of the scattered beam is calculated and is plotted in a graph against the scattering angle.
4. It is seen for an angle of  $50^\circ$  and voltage of 54V, the intensity is maximum.
5. The incident angle  $\theta = 90^\circ - \frac{\phi}{2} = 65^\circ$ . Also, the planar distance for a nickel crystal,  $d = 0.91\text{\AA}$  is known.
6. Using de-Broglie's Wavelength Formula:  $\lambda = \frac{h}{\sqrt{2meV}}$ , we get  $\lambda = 1.66\text{\AA}$
7. Using Bragg's Diffraction Formula:  $\lambda = 2d\sin\theta$ , we get  $\lambda = 1.65\text{\AA}$
8. As these values are nearly identical, we can say that electrons possess the diffraction property, hence proving the wave nature of electrons.

## 1.9 GP Thomson Experiment

1. This experiment also explains the wave nature of electrons.
2. Consider an electron beam, which passes through an anode which deflects the beam into a metal space with a small gap.
3. The electron beam gets incident onto a gold foil of thickness  $0.1 \mu\text{m}$ .
4. After being incident to the gold foil, it gets diffracted onto a screen on which a diffraction pattern is observed.
5. Let the distance between the screen and the foil be  $l$  and the radius of a ring be  $r$ . Also let the incident angle be  $\theta$ .
6. Hence, the diffracted angle will be  $2\theta \implies \tan(2\theta) = \frac{r}{l}$
7. By Bragg's Law,

$$\lambda = 2d\sin\theta \implies d = \frac{\lambda}{2\sin\theta} = \frac{\lambda}{2\theta} = \frac{\lambda}{\tan(2\theta)}$$

Hence,  $\boxed{d = \frac{\lambda l}{r}}$

## 1.10 Some Important Definitions

### 1.10.1 Ideal Waves and Particles:

1. **Ideal Wave:** A non-localized, infinitely long wave with a single frequency.
2. **Ideal Particle:** A localized particle with zero size.

### 1.10.2 Phase Velocity:

Consider an electron with a wave as:  $x = A\cos(kx - \omega t)$ , then  
Phase Velocity,

$$v_p = \frac{\omega}{k} = \frac{\omega\lambda}{2\pi} \implies \boxed{v_p = \nu\lambda}$$

### 1.10.3 Group Velocity:

1. Consider an electron emitting two waves as:  $y_1 = A\cos(k_1x - \omega_1t)$  and  $y_2 = A\cos(k_2x - \omega_2t)$

$$\begin{aligned} \implies y &= y_1 + y_2 = A\cos(k_1x - \omega_1t) + A\cos(k_2x - \omega_2t) \\ \implies y &= 2A\cos\left(\frac{k_1x - \omega_1t + k_2x - \omega_2t}{2}\right)\cos\left(\frac{k_1x - \omega_1t - k_2x + \omega_2t}{2}\right) \\ \implies y &= 2A\cos\left(\frac{(\Delta k)x - (\Delta\omega)t}{2}\right)\cos\left(\frac{(k_1 + k_2)x - (\omega_1 + \omega_2)t}{2}\right) \end{aligned}$$

2. The group velocity is the average of all the individual wave velocities emitted by the electron.
3. We know,  $E = h\nu = mc^2 \implies \nu = \frac{mc^2}{h}$

$$v_p = \nu\lambda = \left(\frac{mc^2}{h}\right)\left(\frac{h}{mv}\right) \implies \boxed{v_p = \frac{c^2}{v}}$$

Hence, minimum value of  $v_p$  is  $c \implies \boxed{v_p \geq c}$

4. Group Velocity,  $\boxed{v_g = \frac{d\omega}{dk}}$

#### 1.10.4 Relation between phase and group velocity:

1. If particle is moving with non-relativistic speed ( $v \ll c$ ):

$$E = \frac{p^2}{2m} \implies \frac{dE}{dp} = \frac{2m}{2p} = v$$

Also,  $p = \hbar k \implies dp = \hbar dk$  and  $E = \hbar\omega \implies dE = \hbar d\omega$

$$\implies \frac{dE}{dp} = \frac{d\omega}{dk} = v_g \implies \boxed{v = v_g}$$

2. If particle is moving with relativistic speed ( $v \approx c$ ):

$$E = p^2c^2 + m_0^2c^4 \implies 2EdE = 2pc^2dp \implies \frac{dE}{dp} = \frac{pc^2}{E}$$

Now,

$$E = mc^2 = \frac{m_0c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \text{ and } p = mv = \frac{m_0v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\implies \frac{dE}{dp} = \frac{(m_0v)c^2}{m_0c^2} = v \implies \boxed{v_g = v}$$

3. If  $\frac{d\omega}{dk}$  is independent of  $k$ , then  $k$  is called "dispersionless medium".

### 1.11 Heisenberg's Uncertainty Principle:

#### 1.11.1 Wave Packets:

1. In classical physics, an electron is just a particle. But in quantum physics, an electron is a "wave packet" with a number of waves with some group velocity as well as a particle inside the packet somewhere.
2. If the wave-packet is large, the velocity can be **fairly** measured accurately, but position of the particle is **very uncertain**.

3. If the wave-packet is small, the position can be **more or less fixed**, but the velocity spread of the packet is large.
4. If the wave-packet is infinitely large, the velocity is **certain** but the position becomes **uncertain**.
5. If the wave-packet is infinitely small, the position is **certain** but the velocity becomes **uncertain**.

### 1.11.2 Statement:

"The product of the uncertainty of position and the uncertainty of momentum of a particle at a given instant is greater than or equal to  $\hbar/2$ ."

$$\boxed{\Delta x \cdot \Delta p \geq \frac{\hbar}{2}} \implies \boxed{\Delta x \cdot \Delta p \geq \frac{h}{4\pi}}$$

### 1.11.3 Proof:

$$v_g = \frac{d\omega}{dk} = \frac{d\omega}{d(\frac{2\pi}{\lambda})} \implies \boxed{v_g = \frac{-\lambda^2 d\omega}{2\pi d\lambda}}$$

$$v_g = \frac{-\lambda^2 d\omega}{2\pi d(\frac{h}{p})} = \frac{-(\frac{h}{p})^2 p^2 d\omega}{-2\pi h dp} = \frac{h d\omega}{-2\pi dp} \implies \boxed{v_g = \frac{\hbar d\omega}{2dp}}$$

$$\text{Also, } v = \frac{\Delta x}{\Delta t} \text{ and } v = v_g \implies \frac{\Delta x}{\Delta t} = \frac{\hbar \Delta \omega}{2 \Delta p} \implies \Delta x \cdot \Delta p = \frac{1}{2} \hbar \Delta \omega \Delta t$$

$$\text{Now, } \Delta t \leq \frac{1}{\Delta \omega} \implies \Delta t \Delta \omega \leq 1$$

$$\implies \boxed{\Delta x \Delta p \geq \frac{1}{2} \hbar} \implies \boxed{\Delta E \Delta t \geq \frac{1}{2} \hbar} \implies \boxed{\Delta \theta \Delta J \geq \frac{1}{2} \hbar}$$

### 1.11.4 Non-Existence of electrons in the nucleus:

The diameter of nucleus of any atom is in the order of  $10^{-14}m$ . If any electron is confined within the nucleus, then the uncertainty of its position  $\Delta x$  must not be greater than  $10^{-14}m$ .

Now,

$$\Delta x \Delta p \geq \frac{h}{2\pi} \implies \Delta p \geq \frac{h}{2\pi \Delta x} = \frac{6.63 \times 10^{-34}}{2(3.14)(10^{-14})} = 1.05 \times 10^{-20} \text{ kgm/s}$$

#### 1. For Non-Relativistic Speed ( $v \ll c$ ):

$$\Delta E = \frac{(\Delta p)^2}{2m} = \frac{(1.05 \times 10^{-20})^2}{(2)(9.1 \times 10^{-31})} = 6.05 \times 10^{-11} \text{ J} = 378 \text{ MeV}$$

The above value of E indicates that an electron with a momentum of  $p = 1.05 \times 10^{-20} \text{ kgm/s}$  and mass of  $m = 9.1 \times 10^{-31} \text{ kg}$  to exist within

the nucleus must have energy  $E = 238MeV$ . But, experimental results of  $\beta$ -decay showed that electron can have a maximum of 3-4 MeV energy. Hence, free electron can't exist within the nucleus.

## 2. For Relativistic Speed ( $v = c$ ):

According to Einstein's Theory of Relativity, Energy is:

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \implies E^2 = \frac{m_0^2 c^6}{c^2 - v^2} \quad (1)$$

Also,

$$p = mv = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}} \implies p^2 = \frac{m_0^2 v^2 c^4}{c^2 - v^2}$$

Multiplying  $c^2$  to both sides,

$$p^2 c^2 = \frac{m_0^2 v^2 c^6}{c^2 - v^2}$$

Subtracting from eq(1),

$$E^2 - p^2 c^2 = \frac{m_0^2 c^6}{c^2 - v^2} - \frac{m_0^2 v^2 c^6}{c^2 - v^2} = \frac{m_0^2 c^4}{c^2 - v^2} (c^2 - v^2)$$

Hence,  $\boxed{E^2 = p^2 c^2 + m_0^2 c^4}$

$$\therefore E^2 = (1.05 \times 10^{-14})^2 (3 \times 10^8)^2 + (9.1 \times 10^{-31})^2 (3 \times 10^8)^4 = 9.929 \times 10^{-24} J$$

$$\implies E = 3.151 \times 10^{-12} J = 19.69 MeV$$

The above value of E indicates that an electron with a momentum of  $p = 1.05 \times 10^{-20} kgm/s$  and mass of  $m = 9.1 \times 10^{-31} kg$  to exist within the nucleus must have energy  $E = 19.69 MeV$ . But, experimental results of  $\beta$ -decay showed that electron can have a maximum of 3-4 MeV energy. Hence, free electron can't exist within the nucleus.

### 1.11.5 Existence of Protons, Neutrons, $\alpha$ -particles in the nucleus:

We know that  $\Delta p = 1.05 \times 10^{-20} kgm/s$

$$E_{p,n} = \frac{p^2}{2m} = \frac{(1.05 \times 10^{-20})^2}{2(1.67 \times 10^{-27})} = 3.3 \times 10^{-14} J = 206 keV$$

And  $E_\alpha = \frac{1}{4} E_p = 51.5 keV$

### 1.11.6 Minimum Energy of a Harmonic Oscillator:

We know,

$$\Delta x \Delta p = \frac{\hbar}{2} \implies \Delta p = \frac{\hbar}{2\Delta x}$$

Now,

$$TE = KE + U = \frac{(\Delta p)^2}{2m} + \frac{1}{2}k(\Delta x)^2$$

$$\therefore TE = \frac{1}{2m} \left( \frac{\hbar}{2\Delta x} \right)^2 + \frac{1}{2}k(\Delta x)^2 = \frac{\hbar^2}{8m(\Delta x)^2} + \frac{1}{2}k(\Delta x)^2$$

For minimum energy,  $(TE)' = 0$

$$-\frac{2\hbar^2}{8m(\Delta x)^3} + \frac{1}{2}k(2\Delta x) = 0 \implies \Delta x = \left( \frac{\hbar^2}{4mk} \right)^{1/4}$$

$$\therefore TE = \frac{\hbar^2}{8m \left( \frac{\hbar^2}{4mk} \right)^{1/2}} + \frac{1}{2}k \left( \frac{\hbar^2}{4mk} \right)^{1/2} \implies TE = \frac{\hbar}{2} \sqrt{\frac{k}{m}} = \frac{1}{2}\hbar\omega$$

### 1.11.7 Determination of the size of an atom:

According to Heisenberg's uncertainty principle,  $\Delta p \Delta x = \hbar$

Let the size of an atom be  $a \implies a \Delta p = \hbar \implies \Delta p = \hbar/a$  Now,

$$TE = KE + U = \frac{p^2}{2m} - \frac{e^2}{a} \implies TE = \frac{\hbar^2}{2ma^2} - \frac{e^2}{a}$$

For minimum energy,  $(TE)' = 0$

$$\therefore \frac{\hbar^2}{ma^3} = \frac{e^2}{a^2} \implies a = \frac{\hbar^2}{e^2 m}$$

On substituting the values of  $\hbar$ ,  $e$ ,  $m$ , we get  $a = 0.53A$

Similarly, we get  $TE = 13.6eV$

## 1.12 Wave Function:

1. In the case of matter waves, the quantity that varies periodically is called the "wave function" ( $\psi$ ).
2. Probability / Density of Wave Function ( $P$ ) =  $\psi \cdot \psi^*$   
where  $\psi^*$  is the complex conjugate of  $\psi$ .
3.  $\int_{-\infty}^{\infty} \psi \cdot \psi^* d\tau = 1$  where  $d\tau = dx \cdot dy \cdot dz$
4. **Properties of Wave Function:**
  - (a) It must be finite.
  - (b) It must have a single value.
  - (c) It must be continuous and have a confined first order derivation.

### 1.12.1 Linearity and Superposition:

Consider an electron wave is incident on a plane with two slits, and observe the interference pattern.

1. If both slits are closed, the probability density,  $P_0 = 0$
2. If only slit 1 is open, the probability density,  $P_1 = |\psi_1|^2 = \psi_1 \cdot \psi_1^*$
3. If only slit 2 is open, the probability density,  $P_2 = |\psi_2|^2 = \psi_2 \cdot \psi_2^*$
4. If both slits are closed, the probability density,  $P_3 = P_1 + P_2 = |\psi_1 + \psi_2|^2$

$$\therefore P_3 = (\psi_1 + \psi_2)(\psi_1^* + \psi_2^*) = \psi_1\psi_1^* + \psi_2\psi_2^* + \psi_1\psi_2^* + \psi_2\psi_1^*$$

Hence,

$$\boxed{P_3 = P_1 + P_2 + \psi_1\psi_2^* + \psi_2\psi_1^*}$$

where,  $\psi_1\psi_2^* + \psi_2\psi_1^*$  is the oscillation of electron intensity.

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### 1.12.2 Normalized and Orthogonal Functions:

1.  $\int_{-\infty}^{\infty} \psi_m \cdot \psi_n^* d\tau = \delta_{mn} = 1$  (for  $m = n$ ) means that an electron will be 100% in the given wave packet. This is called the **Normalized Function**.
2.  $\int_{-\infty}^{\infty} \psi_m \cdot \psi_n^* d\tau = \delta_{mn} = 0$  (for  $m \neq n$ ) means that an electron will be 100% NOT in the given wave packet. This is called the **Orthogonal Function**.

### 1.12.3 Wave Equation:

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}}$$

1. **Question:** Check if  $\psi = A \cos(\omega t - kx)$  satisfies the wave equation.

$$\frac{\partial \psi}{\partial x} = k A \sin(\omega t - kx) \implies \frac{\partial^2 \psi}{\partial x^2} = k^2 A \cos(\omega t - kx) = k^2 \psi$$

$$\frac{\partial \psi}{\partial t} = \omega A \sin(\omega t - kx) \implies \frac{\partial^2 \psi}{\partial t^2} = \omega^2 A \cos(\omega t - kx) = \omega^2 \psi$$

But we know,  $\omega = kv \implies \omega^2 = k^2 v^2$

$$\therefore \frac{\partial^2 \psi}{\partial t^2} = v^2 k^2 \psi = v^2 \left( \frac{\partial^2 \psi}{\partial x^2} \right) \implies \boxed{\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}}$$

Hence, the given wave function satisfies the wave equation.

2. **Question:** Check if  $\psi = Ae^{i\omega(t-x/v)}$  satisfies the wave equation.

$$\frac{\partial \psi}{\partial x} = \frac{-i\omega}{v} Ae^{i\omega(t-x/v)} \implies \frac{\partial^2 \psi}{\partial x^2} = \left(\frac{i\omega}{v}\right)^2 Ae^{i\omega(t-x/v)} = \frac{-\omega^2}{v^2} \psi$$

$$\frac{\partial \psi}{\partial t} = -i\omega Ae^{i\omega(t-x/v)} \implies \frac{\partial^2 \psi}{\partial t^2} = (i\omega)^2 Ae^{i\omega(t-x/v)} = -\omega^2 \psi$$

Putting the result in the first equation,

$$\implies \boxed{\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}}$$

Hence, the given wave function satisfies the wave equation.

## 1.13 Schrödinger's Wave Equation

### 1.13.1 Schrödinger's Time Independent Wave Equation:

Consider a system of stationary waves to be associated with a particle, and  $\psi(r, t)$  be the characteristic wave equation.

where  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$  is the position of the particle and  $t$  is the time.

The differential equation of wave motion in 3-D in accordance with Maxwell's wave equation is:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = \nabla^2 \psi \quad (1)$$

$\nabla^2$  is called the "Laplace Operator".

The solution of eq(1) gives  $\psi$  as a periodic displacement in terms of  $\psi$  as:

$$\psi(r, t) = \psi_0(r, t)e^{-i\omega t} \quad (2)$$

Differentiating eq(2) wrt  $t$ ,

$$\frac{\partial \psi}{\partial t} = -i\omega e^{-i\omega t}$$

Differentiating again wrt  $t$ ,

$$\frac{\partial^2 \psi}{\partial t^2} = -\omega^2 e^{-i\omega t} = -\omega^2 \psi = -(2\pi\nu)^2 \psi = -\left(2\pi \frac{v}{\lambda}\right)^2 \psi$$

$$\frac{\partial^2 \psi}{\partial t^2} = -\frac{4\pi^2 v^2}{\lambda^2} \psi$$

From eq(1), we get

$$\nabla^2 \psi = \frac{1}{v^2} \cdot \frac{4\pi^2 v^2}{\lambda^2} \psi = -\frac{4\pi^2}{\lambda^2} \psi = -\frac{4\pi^2 p^2}{h^2} \psi = -\frac{m^2 v^2}{h^2} \psi$$

Hence,

$$\nabla^2 \psi + \frac{m^2 v^2}{h^2} \psi = 0$$

If E and V are the total and potential energies of the particle respectively, then its KE = E - V =  $\frac{1}{2}mv^2 \implies m^2v^2 = 2m(E - V)$

Hence,

$$\boxed{\nabla^2\psi + \frac{2m(E - V)}{\hbar^2}\psi = 0} \quad - (3)$$

This is the "Schrödinger Time Independent Wave Equation".

For freely moving particle, V = 0,

$$\boxed{\nabla^2\psi + \frac{2mE}{\hbar^2}\psi = 0}$$

### 1.13.2 Schrödinger's Time Dependent Wave Equation:

In order to obtain the time dependent equation, we eliminate the total energy E from the time independent equation.

We know that, from eq(2),

$$\frac{\partial\psi}{\partial t} = -i(2\pi\nu)\psi = -i(2\pi\frac{E}{h})\psi$$

Multiplying and dividing by i,

$$\frac{\partial\psi}{\partial t} = \frac{-i^2 2\pi E\psi}{ih} = \frac{E\psi}{i\hbar} \implies E\psi = i\hbar \frac{\partial\psi}{\partial t} \quad - (4)$$

From eq(3),

$$\nabla^2\psi + \frac{2m(E - V)}{\hbar^2}\psi = 0 \implies \nabla^2\psi + \frac{2m}{\hbar^2}(E\psi - V\psi) = 0$$

From eq(4),

$$\begin{aligned} \therefore \nabla^2\psi + \frac{2m}{\hbar^2}(i\hbar \frac{\partial\psi}{\partial t} - V\psi) &= 0 \implies -\frac{\hbar^2}{2m}\nabla^2\psi = i\hbar \frac{\partial\psi}{\partial t} - V\psi \\ -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi &= i\hbar \frac{\partial\psi}{\partial t} \implies \boxed{\left[-\frac{\hbar^2}{2m}\nabla^2 + V\right]\psi = i\hbar \frac{\partial\psi}{\partial t}} \end{aligned}$$

This is the "Schrödinger Time Dependent Wave Equation".

It can also be written as:  $\boxed{H\psi = E\psi}$

where  $H = -\frac{\hbar^2}{2m}\nabla^2 + V$  is called as the "Hamiltonian Operator".  
and  $E = i\hbar \frac{\partial}{\partial t}$  is called as the "Energy Operator".

## 1.14 Applications of Schrödinger's Equations:

### 1.15 Infinite Potential Well:

Consider a 1-D infinite depth well of width L (x from 0 to L).

The potential energy of a particle is given as:

$$\begin{aligned} V &= 0 & 0 \leq x \leq L \\ V &= \infty & x \leq 0, x \geq L \end{aligned}$$

The Schrödinger Time Independent Wave Equation in 1-D is:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2mE}{\hbar^2} \psi = 0 \implies \frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0$$

where

$$k^2 = \frac{2mE}{\hbar^2} \implies k = \frac{\sqrt{2mE}}{\hbar}$$

The general solution of above equation is:  $\psi(x) = A \sin(kx) + B \cos(kx)$

Applying Boundary Condition ( $x = 0$ ):  $0 = A \sin(0) + B \cos(0) \implies B = 0$

Applying Boundary Condition ( $x = L$ ):  $0 = A \sin(kL) \implies kL = n\pi$

$$k^2 = \frac{2mE}{\hbar^2} = \frac{n^2 \pi^2}{L^2} \implies \boxed{E = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 h^2}{8mL^2}}$$

From the above equation we can conclude that in an infinite potential well, the particle cannot have arbitrary energy, but can take only certain energy values corresponding to  $n = 1, 2, 3, \dots$

Now, the probability of finding a particle within the well is:

$$P = \int_0^L \psi \cdot \psi^* dx = 1$$

Also,

$$\psi = \psi^* = A \sin\left(\frac{n\pi}{L}x\right) \implies P = \int_0^L A^2 \sin^2\left(\frac{n\pi}{L}x\right) dx = 1$$

On Solving we get,

$$A = \sqrt{\frac{2}{L}} \implies \boxed{\psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right)}$$

### 1.16 Finite Potential Step:

Consider a finite depth step of height  $V_0$  and particle energy  $E$  such that  $E > V_0$ . The potential energy of a particle is given as:

$$\begin{aligned} V &= V_0 & x > 0 \\ V &= 0 & x < 0 \end{aligned}$$

Lets consider the left part of  $x = 0$  as Region 1 and the other Region 2.

We consider that when a particle (energy  $E$ ) is incident from left onto potential step of height  $V_0$  there should be no reflection at the boundary ( $x = 0$ ) according to classical physics. However in quantum mechanics, it is not true. It means that there will be some reflection from the boundary of potential step.

$$\lambda_1 = \frac{h}{p_1} = \frac{h}{\sqrt{2mE}}, \lambda_2 = \frac{h}{p_2} = \frac{h}{\sqrt{2m(E - V_0)}}$$

The wavelength of particle suddenly changes from region 1 to 2. Hence a small part of the wave associated with the particle is reflected due to the change in wavelength and the rest is transmitted.

In Region 1,

$$\frac{\partial^2 \psi_1(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_1(x) = 0$$

In Region 2,

$$\frac{\partial^2 \psi_2(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_2(x) = 0$$

Let,

$$k_1^2 = \frac{2mE}{\hbar^2} \implies \frac{\partial^2 \psi_1(x)}{\partial x^2} + k_1^2 \psi_1(x) = 0$$

$$k_2^2 = \frac{2m(E - V_0)}{\hbar^2} \implies \frac{\partial^2 \psi_2(x)}{\partial x^2} + k_2^2 \psi_2(x) = 0$$

The solutions of the above equations are:

$$\psi_1(x) = A_1 e^{ik_1 x} + A_2 e^{-ik_1 x}$$

$$\psi_2(x) = A_3 e^{ik_2 x} + A_4 e^{-ik_2 x}$$

where  $A_1, A_2, A_3, A_4$  are constants.

The first term in first equation represents the wave traveling in the positive direction in the first region whereas the 2nd term represents the reflected part of incident wave in the first region.

The first term in the second equation represents the transmitted part of the incident wave in the 2nd region whereas the second term has no meaning as there is no reflection in the 2nd region.

$$\therefore \psi_2(x) = A_3 e^{ik_2 x}$$

Boundary Condition:

$$\psi_1(x)|_{x=0} = \psi_2(x)|_{x=0} \implies A_1 + A_2 = A_3$$

$$\left. \frac{\partial \psi_1(x)}{\partial x} \right|_{x=0} = \left. \frac{\partial \psi_2(x)}{\partial x} \right|_{x=0} \implies k_1 A_1 + k_1 A_2 = k_2 A_3$$

From above two equations,

$$\frac{A_3}{A_1} = \frac{2k_1}{k_1 + k_2}$$

Since the reflection and transmission of particles takes place, the problem can be investigated based on the coefficient  $A_1$  related to  $\psi_1$  (incident),  $A_2$  related to  $\psi_1$  (reflected), and  $A_3$  related to  $\psi_2$  (transmitted).

Transmission Coefficient:

$$T = \frac{I_3}{I_1} = \left( \frac{A_3}{A_1} \right)^2 = \left( \frac{2k_1}{k_1 + k_2} \right)^2$$

Reflection Coefficient:

$$R = 1 - T = \left( \frac{k_1 - k_2}{k_1 + k_2} \right)^2$$

### 1.17 Tunelling Effect

Consider a tunnel from  $x = 0$  to  $x = a$  with potential given as:

$$\begin{aligned} V(x) &= 0 & x < 0 & \quad (Region - 1) \\ V(x) &= V_0 & 0 < x < a & \quad (Region - 2) \\ V(x) &= 0 & x > a & \quad (Region - 3) \end{aligned}$$

The particle incident to the barrier of region 2 has an energy  $E$  which is less than the barrier height  $V_0$ . According to classical mechanics, particle can never penetrate the barrier and appear in region 3 ( $R=1$ ) i.e. particle is always reflected from the barrier ( $T=0$ ). But according to quantum mechanics, this is not true and there is some probability to penetrate the barrier. This means that a fraction of the particle will cross the barrier and appear in region 3 ( $T \neq 0$ ,  $R \neq 1$ ). The probability of crossing the barrier is called "tunneling effect".

Schrödinger's equations for Regions 1-3 are:

$$\begin{aligned} \frac{\partial^2 \psi_1(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_1(x) &= 0 \implies \frac{\partial^2 \psi_1(x)}{\partial x^2} + k_1^2 \psi_1(x) = 0 \\ \frac{\partial^2 \psi_2(x)}{\partial x^2} - \frac{2m(E - V_0)}{\hbar^2} \psi_2(x) &= 0 \implies \frac{\partial^2 \psi_2(x)}{\partial x^2} - k_2^2 \psi_2(x) = 0 \\ \frac{\partial^2 \psi_3(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_3(x) &= 0 \implies \frac{\partial^2 \psi_3(x)}{\partial x^2} + k_1^2 \psi_3(x) = 0 \end{aligned}$$

where

$$k_1^2 = \frac{2mE}{\hbar^2}, \quad k_2^2 = \frac{2m(E - V_0)}{\hbar^2}$$

The solutions to the above equations are:

$$\begin{aligned} \psi_1(x) &= A_1 e^{ik_1 x} + A_2 e^{-ik_1 x} \\ \psi_3(x) &= A_3 e^{ik_1 x} + A_4 e^{-ik_1 x} \\ \psi_2(x) &= A_5 e^{-k_2 x} + A_6 e^{k_2 x} \end{aligned}$$

where  $A_i$  is constant. As no wave is reflected in region-3,  $\psi_3(x) = A_3 e^{ik_1 x}$

Boundary Condition ( $x=0$ ):

$$\begin{aligned} \psi_1(x)|_{x=0} &= \psi_2(x)|_{x=0} \implies A_1 + A_2 = A_5 + A_6 \\ \left. \frac{\partial \psi_1(x)}{\partial x} \right|_{x=0} &= \left. \frac{\partial \psi_2(x)}{\partial x} \right|_{x=0} \implies ik_1 A_1 + ik_1 A_2 = -k_2 A_5 + k_2 A_6 \end{aligned}$$

Boundary Condition( $x=a$ ):

$$\begin{aligned} \psi_3(x)|_{x=a} &= \psi_2(x)|_{x=a} \implies A_3 e^{ik_1 a} = A_5 e^{-k_2 a} + A_6 e^{k_2 a} \\ \left. \frac{\partial \psi_3(x)}{\partial x} \right|_{x=a} &= \left. \frac{\partial \psi_2(x)}{\partial x} \right|_{x=a} \implies ik_1 A_3 e^{ik_1 a} = -k_2 A_5 e^{-k_2 a} + k_2 A_6 e^{k_2 a} \end{aligned}$$

On Solving the above equations,

$$A_1 = \frac{ik_1 - k_2}{2ik_1} A_5 + \frac{ik_1 + k_2}{2ik_1} A_6$$
$$A_6 = \frac{k_2 + ik_1}{2k_2 e^{k_2 a}} e^{ik_1 a} A_3; \quad A_5 = \frac{k_2 - ik_1}{2k_2 e^{k_2 a}} e^{ik_1 a} A_3$$

## 2 Modern Energy Sources

### 2.1 Introduction

1. **Symbolic Representation of an Element:**  ${}_A^ZX$   
where Z = atomic number = number of protons  
A = atomic mass number and A - Z = number of neutrons.
2. **Isotopes:** Elements with the same atomic number, but different atomic mass number.  
Example,  ${}_1^1H$ ,  ${}_1^2H$ ,  ${}_1^3H$  are isotopes of Hydrogen.

#### 3. Isotopes of Uranium:

| Isotope          | % Abundance | Half-life   |
|------------------|-------------|-------------|
| ${}_{92}^{238}U$ | 99.28%      | 4.5e9 years |
| ${}_{92}^{235}U$ | 0.714%      | 7.1e8 years |
| ${}_{92}^{234}U$ | 0.006%      | 2.5e6 years |

4. **Proton-Neutron Theory:** For a stable atom, the number of neutrons and protons are equal when the atomic number is less than 20. For atoms with atomic number more than 20, the neutrons are more than protons for stable atom.
5. **Mass of 1amu in kg:**

$$1amu = \frac{1}{12} \times m(C-12) = 1.66 \times 10^{-27}kg$$

6. **Mass of  ${}_{92}^{238}U$  atom:**  $238amu = 238 \times 1.66 \times 10^{-27}kg = 3.95 \times 10^{-25}kg$
7. **Atomic Radius:**  $R = R_0A^{1/3}$  where  $R_0 = 1.4 \times 10^{-15}m$
8. **Nuclear Density:**

$$\rho = \frac{M}{V} = \frac{M}{4/3\pi R^3} = \frac{3M}{4\pi R_0^3 A} = \frac{3 \times (1.66 \times 10^{-27})}{4\pi R_0^3}$$

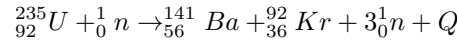
Hence,  $\rho = 1.66 \times 10^{17}kg/m^3$   
Nuclear Density is same for all atoms.

9. **Mass Defect:**  $\Delta m = (Zm_p + (A - Z)m_n) - M$   
where  $m_p$  = mass of proton,  $m_n$  = mass of neutron, M = mass of nucleus.
10. **Binding Energy**  $\Delta E = \Delta mc^2$  where  $c = 931.4MeV/amu$   
**Question:** Given that mass of nucleus of  ${}_{7}^{14}N$  is 14.00307amu,  $m_p = 1.00783amu$ , and  $m_n = 1.00867amu$ , calculate the binding energy of  ${}_{7}^{14}N$ .  
 $\Delta m = (7 \times 1.00783 + 7 \times 1.00867) - 14.00307 = 0.11243amu$   
 $\therefore \Delta E = \Delta mc^2 = (0.11243)(931.4) = 104.717MeV$

## 2.2 Nuclear Fission:

1. The process of splitting a heavy atom nucleus into two smaller atoms of roughly same size by collision of a neutron is called "nuclear fission".
2. **Slow Neutrons:** Atoms with odd atomic mass number split by slow neutrons. Slow neutrons possess kinetic energy of less than 100keV.
3. **Slow Neutrons:** Atoms with even atomic mass number split by fast neutrons. Fast neutrons possess kinetic energy of 100keV to 50MeV.

### 2.2.1 Nuclear Fission by U-235:



1. Before fission reaction,  
 $m_1 = 235.125 + 1.009 = 236.134\text{amu}$
2. After fission reaction,  
 $m_2 = 140.958 + 91.926 + 3(1.009) = 235.911\text{amu}$
3. Hence, mass defect,  $\Delta m = 236.124 - 235.911 = 0.223\text{amu}$   
 $\therefore \Delta E = \Delta mc^2 = (0.223)(931.4) = 207.7022\text{MeV} = 3.2 \times 10^{-11}\text{J}$
4. Due to 1gm Uranium,  
 $\text{energy produced} = 3.2 \times 10^{-11} \times \left(\frac{6.023 \times 10^{23}}{235}\right) = 8.516 \times 10^{10}\text{J}$   
 $= 23.655\text{MWh} = 2.033 \times 10^{10}\text{cal}$
5. For burning coal, 1kg coal = 7000 kcal/kg  
Hence, to get the same amount of energy as 1gm uranium, amount of coal required =

$$\frac{2 \times 10^{10}}{7000 \times 10^3} = 2857\text{kg} \approx 3000\text{kg}$$

6. **Question:** A city requires 100 MW power on average if efficiency of nuclear reaction is 20% by using U-235 as nuclear fuel calculate how much fuel is required.

$$\text{Energy per day} = 100 \times 10^6 \times 86400 = 8.64 \times 10^{12}\text{J}$$

Let 'm' kg of fuel is required.

$$\text{Number of atoms required} = m \times \left(\frac{6.023 \times 10^{26}}{235}\right) = m \times (2.562 \times 10^{24})$$

$$\text{Energy} = N\eta\Delta E = m(2.562 \times 10^{24})(0.2)(3.2 \times 10^{-11}) = m(1.6416 \times 10^{13})$$

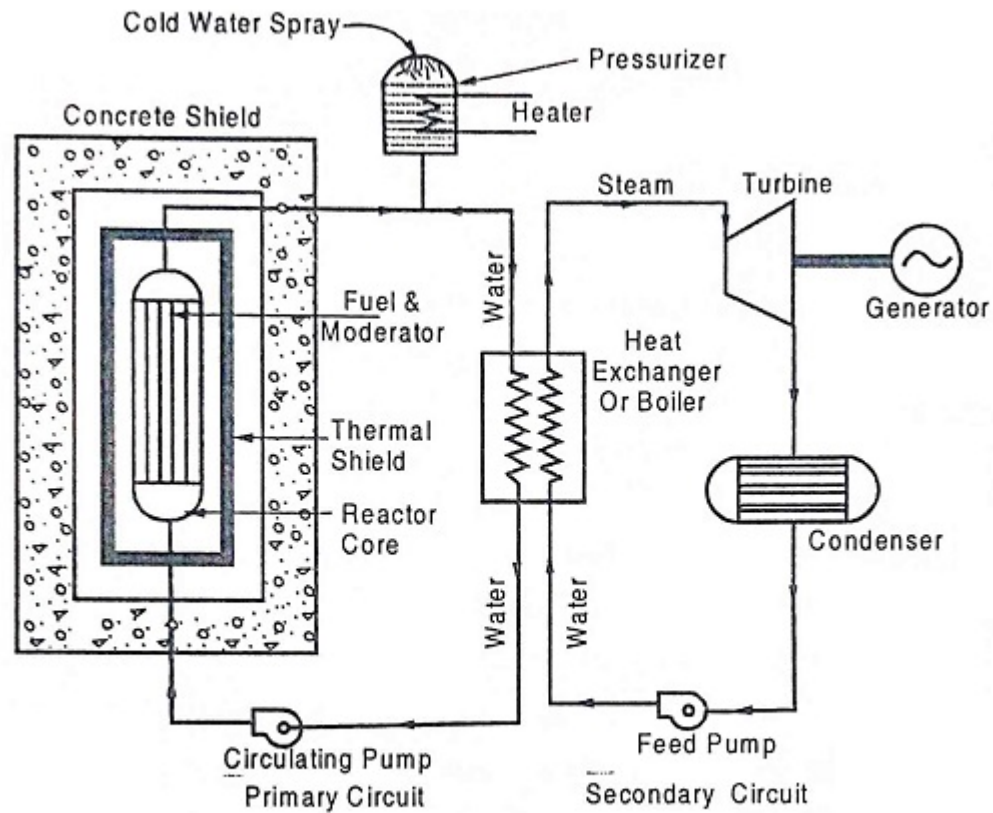
$$\therefore m(1.6416 \times 10^{13}) = 8.64 \times 10^{12} \implies m = 0.526\text{kg}$$

## 2.3 Nuclear Reactor:

1. **Fuel:** Availability of U-235 is 0.71%, whereas of U-238 is 99.28%.  
For making cheaper fuel, 1 atom of U-235 is added to 140 atoms of U-238.  
This is called as "**enrichment proces**".  
 $U - 238 + {}^1_0n \rightarrow Pu - 239$ . This is called as "**breeding process**".
2. **Moderator:** Fast neutrons doesn't react with U-235. Hence, to reduce the speed of these neutrons, moderators like water, heavy water, graphite, beryllium are used.
3. **Reflectors:** These reflect escaping neutrons back to the reactor core. Thorium and Nickel are used as reflectors.
4. **Coolant:** These are used to cool down the reactor core and transfer excess heat to the thermal exchange units. Examples, Water, liquid metals.
5. **Control Rods:** These are made of Boron or Cadmium. These absorb the neutrons either completely or partially.
6. **Structural / Cladding Materials:** These are used to make the reactor itself. Examples, Zirconium, Titanium.
7. **Reactor Shielding:** Thermal shielding is used to control heat, made by steel walls. Biological shielding is used to control radioactivity, made by concrete walls.

## 2.4 Types of Nuclear Power Plants:

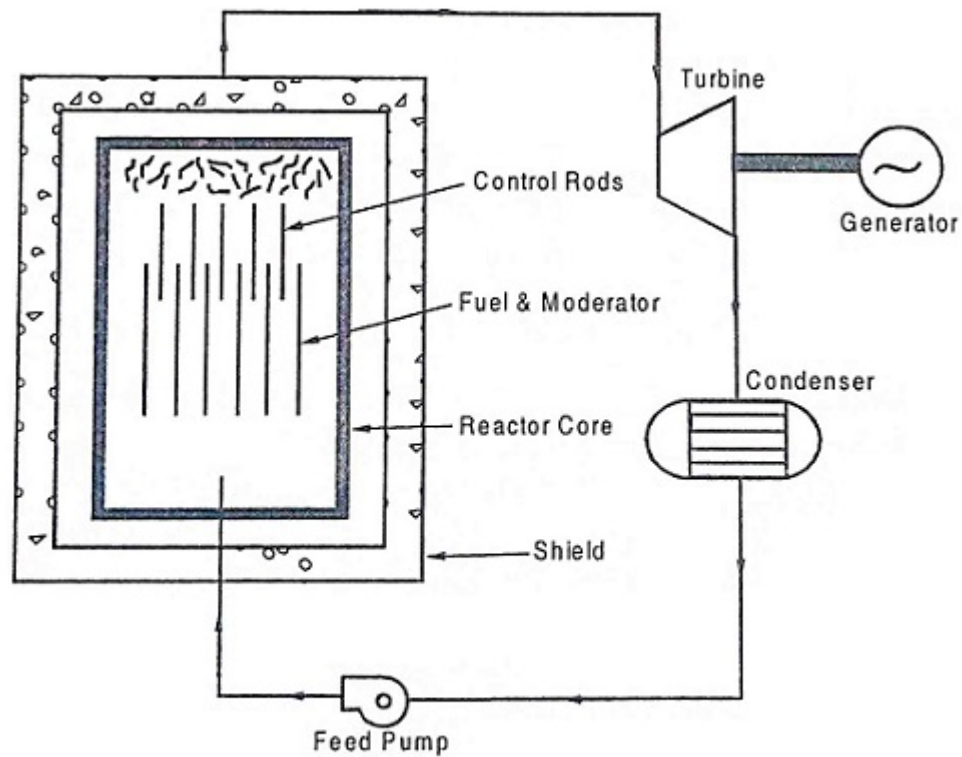
### 2.4.1 Pressurized Water Reactor (PWR):



**Fig:3.7 Pressurized Water Reactor Nuclear Power Plant**

1. It is a type of Thermal Reactor.
2. Enriched Uranium (3% U-235, 97% U-238) is used as fuel.
3. 130-150 bar pressure is produced.
4. Water can be used as both coolant and moderator.
5. Reactor is small in size as it doesn't require much control rods.
6. Fissionable Materials remain in the reactor.
7. It is expensive and the water can get contaminated.
8. Energy efficiency of this reactor is also less.

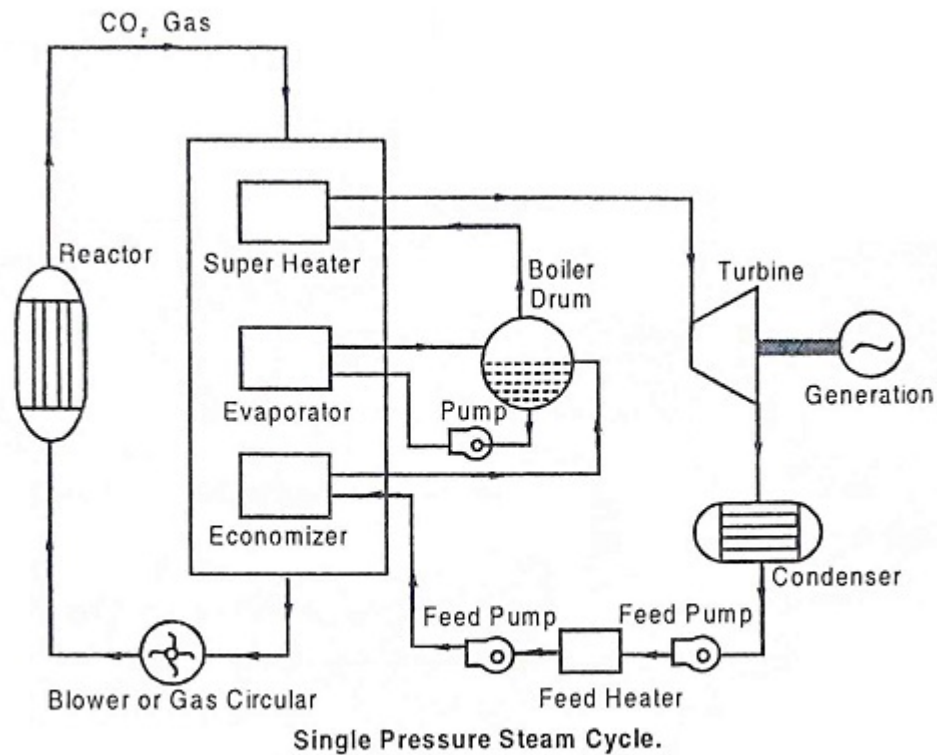
#### 2.4.2 Boiling Water Reactor (BWR):



**Fig:3.6 Boiling Water Reactor**

1. Saturated steam of  $285^{\circ}\text{C}$  at 70 bar is obtained.
2. Enriched Uranium is used as fuel.
3. Water is used as both coolant and moderator
4. Contamination of water has a higher possibility.
5. Energy efficiency is very less.
6. It eliminates the heat exchange unit, pressurizer, pipes, etc. which cuts down the cost of the plant.

### 2.4.3 Gas Cooled Reactor (GCR):

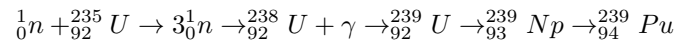


**Fig:3.10 Gas Cooled Reactor**

1. Coolant used is CO<sub>2</sub> gas or He gas, which is less contaminated than water.
2. For normal GCR, pressure of 7 bar at 336°C is obtained.
3. For high temperature gas cooled reactor, a pressure of 15-30 bar is obtained at a temperature of 700-800°C.
4. Corrosion will not take place as water is not used here.
5. Graphite is used as a moderator.
6. It requires a large heat exchange unit and is expensive.
7. CO<sub>2</sub>/He gas leakage may take place.
8. More power is used for cooling.

#### 2.4.4 Fast Breeder Type Nuclear Reactors:

##### 1. Breeding Process:



##### 2. Liquid Metal Fast Breeder Reactor:

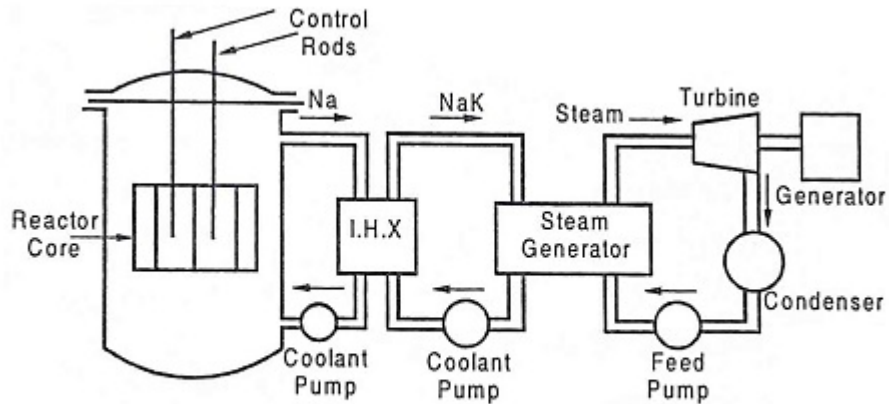


Fig:3.11 Fast Breeder Reactor

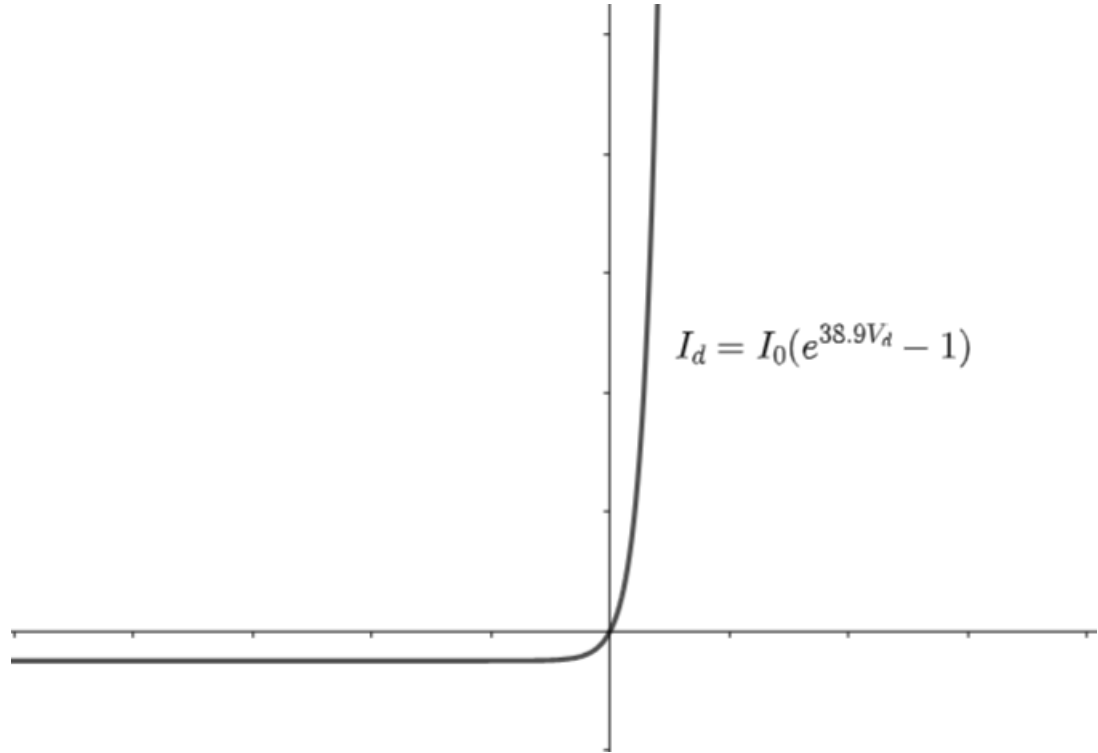
Here liquid sodium is used as coolant and graphite is used as a moderator. It reaches 250° at 7 bar of pressure.

##### 3. Gas Cooled Fast Breeder Reactor: It works similar to the liquid metal reactor, but CO<sub>2</sub> gas is used instead of liquid Na.

#### 2.5 Solar Cell:

It works on the principle of photovoltaic effect.

### 2.5.1 I-V Characteristics of P-N Junction Diode



$$I_d = I_0(\exp(\frac{eV}{kT}) - 1) \implies I_d = I_0(e^{38.9V_d} - 1)$$

**Question:** Let a p-n junction diode at  $25^\circ$  have a saturation current of  $10^{-9}$  A. Find its voltage when it is carrying a current of (a) 0A, (b) 1A, (c) 10A.

**Solution:** We know,

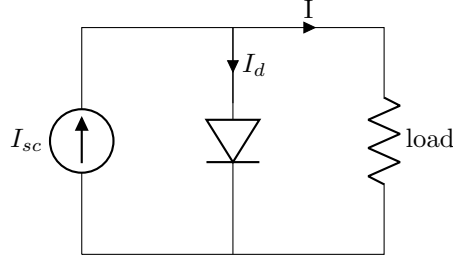
$$V_d = \frac{1}{38.9} \ln \left( 1 + \frac{I_d}{I_0} \right)$$

$$V_d \Big|_{0A} = \frac{1}{38.9} \ln \left( 1 + \frac{0}{I_0} \right) = 0$$

$$V_d \Big|_{1A} = \frac{1}{38.9} \ln \left( 1 + \frac{1}{I_0} \right) = 0.533A$$

$$V_d \Big|_{10A} = \frac{1}{38.9} \ln \left( 1 + \frac{10}{I_0} \right) = 0.592A$$

### 2.5.2 I-V Characteristics of Solar Cell



1.  $I_{sc} \propto$  intensity of light
2. **Short Circuit Current:**  $V = 0, I = I_{sc}$
3. **Open Circuit Voltage:**  $V = V_{oc}, I = 0$
4. By KCL,  $I = I_{sc} - I_d = I_{sc} - I_0(e^{38.9V_{oc}} - 1)$   
Hence,

$$V_{oc} = \frac{1}{38.9} \ln \left( \frac{I_{sc}}{I_0} + 1 \right)$$

5. **Maximum Power:**  $P_{max} = I_R \cdot V_R$
6. **Cell Efficiency:**

$$\eta = \frac{P_{out}}{P_{in}} = \frac{P_{max}}{I \times A} = \frac{P_{max}}{1000A}$$

Radiation Power on earth (I) = 1000W/m<sup>2</sup>.

7. **Fill Factor:**

$$f = \frac{P_{max}}{P_{ideal}} = \frac{I_R V_R}{I_{sc} V_{oc}} \approx 70\%$$

8. **Effect of Temperature on Solar Cells:** When temperature is increased,  $V_{oc}$  decreases and small current increase occurs. Hence, we get more power in lower temperatures.
9. Individual solar cells give about 0.5V voltage and 2-3A current, in a solar cell "module", cells are connected in series as to give 15-18V voltage.

## 3 Optics:

### 3.1 Laser:

**LASER** stands for Light Amplification and Stimulated Emission of Radiation.

1. Laser is monochromatic (same amplitude, frequency, etc.).
2. It is directional.
3. It is highly coherent (same phase).
4. It's intensity is higher than normal light.

### 3.2 Interaction of Light with Matter:

#### 3.2.1 Absorption:

If A is atom of lower energy level, and  $A^*$  is atom of higher energy level, then:

$$A + h\nu \rightarrow A^*$$

Probability of absorption  $\propto$  Photon Density:

$$P_{12} \propto \rho(\nu) \implies P_{12} = B_{12}\rho(\nu)$$

where  $B_{12}$  is called as "Einstein Absorption Coefficient".

Number of Atoms Absorbed:

$$N_{ab} = P_{12}N_1\Delta t = B_{12}\rho(\nu)N_1\Delta t$$

#### 3.2.2 Spontaneous Emission:

$$A^* \rightarrow A + h\nu$$

$$P_{21(sp)} = A_{21}N_2$$

where  $A_{21}$  is called the "Einstein Spontaneous Coefficient"

$$N_{sp} = A_{21}N_2\Delta t$$

#### 3.2.3 Stimulated Emission:

$$A^* + h\nu \rightarrow A + 2h\nu$$

$$P_{21(st)} = B_{21}\rho(\nu)$$

where  $B_{21}$  is called the "Einstein Stimulated Coefficient"

$$N_{st} = B_{21}N_2\rho(\nu)\Delta t$$

### 3.2.4 Relation between Einstein's Coefficients:

The number of atoms absorbing photons per second per unit volume  
 = The number of atoms emitting photons per second per unit volume.

$$N_{ab} = N_{sp} + N_{st}$$

$$B_{12}\rho(\nu)N_1\Delta t = A_{21}N_2\Delta t + B_{21}N_2\rho(\nu)\Delta t$$

$$\therefore B_{12}\rho(\nu)N_1 = A_{21}N_2 + B_{21}N_2\rho(\nu)$$

$$\rho(\nu)[B_{12}N_1 - B_{21}N_2] = A_{21}N_2$$

$$\rho(\nu) = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2} = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2} \frac{B_{12}N_2}{B_{12}N_2}$$

$$\rho(\nu) = \frac{\frac{A_{21}}{B_{12}}}{\frac{N_1}{N_2} - \frac{B_{21}}{B_{12}}}$$

According to Planck's Radiation Law,

$$\rho(\nu) = \frac{8\pi hc}{\lambda^5} \left( \frac{1}{e^{h\nu/kT} - 1} \right)$$

Now,

$$\frac{N_2}{N_1} = e^{-\frac{E_2 - E_1}{kT}} = e^{-h\nu/kT} \implies \frac{N_1}{N_2} = e^{h\nu/kT}$$

Hence,

$$\rho(\nu) = \frac{8\pi hc}{\lambda^5} \left( \frac{1}{e^{h\nu/kT} - 1} \right) = \frac{A_{21}}{B_{12}} \left( \frac{1}{e^{h\nu/kT} - \frac{B_{21}}{B_{12}}} \right)$$

On Comparison,

$$\frac{B_{21}}{B_{12}} = 1 \implies B_{21} = B_{12}$$

and

$$\frac{A_{21}}{B_{12}} = \frac{8\pi hc}{\lambda^5} \implies \boxed{B_{12} = B_{21} = \frac{\lambda^5}{8\pi hc} A_{21}}$$

**Question:** A 10mW He-Ne Laser with efficiency 1%. Assuming all input energy is used in pumping an atom from ground to excited state, which is equal to 20eV. Find how many atoms are promoted.

**Solution:**

$$\eta = \frac{P_{out}}{P_{in}} = \frac{10mW}{P_{in}} \implies P_{in} = \frac{10 \times 10^{-3}}{1 \times 10^{-2}} = 1W$$

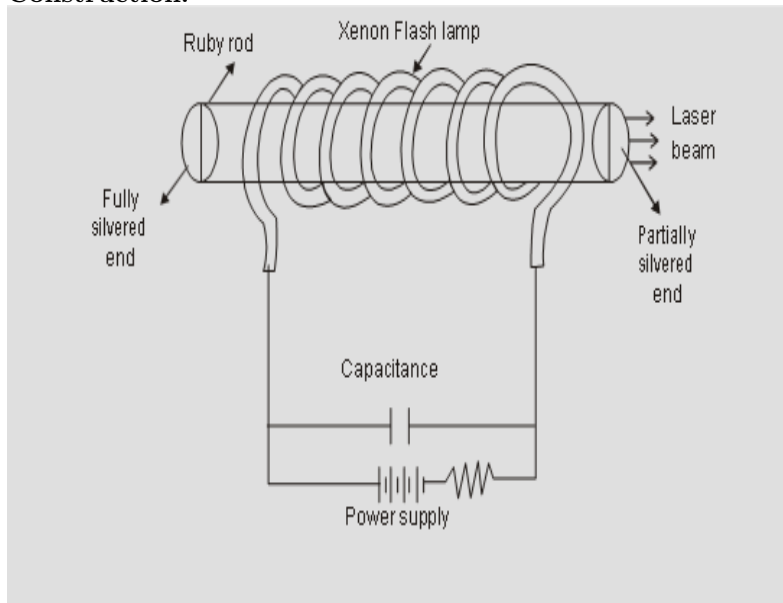
Hence, Number of Atoms,

$$N = \frac{P_{in}}{E} = \frac{1}{20 \times 1.6 \times 10^{-19}} = 3.125 \times 10^{17} atoms$$

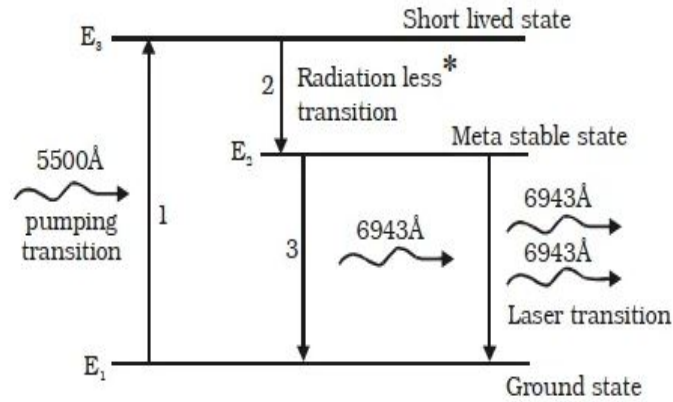
### 3.3 Types of Laser Systems:

#### 3.3.1 Ruby Laser:

1. It is a 3-level laser system.
2. **Composition:**  $N_2O_3$  (99.95%) and  $Cr$  (0.05%).
3. Cr is the active medium. ( $Cr^{+3}$  ions)
4. Only green and blue lights are absorbed.
5. Optical pumping occurs.
6. Light is in pulses.
7. Silver coating is given to the end surfaces of ruby crystal.
8. **Construction:**

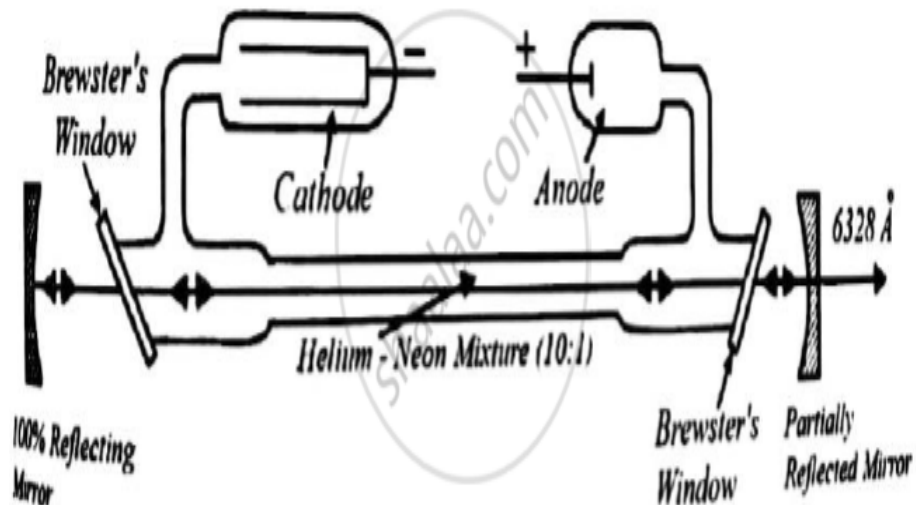


### 9. Energy Diagram:

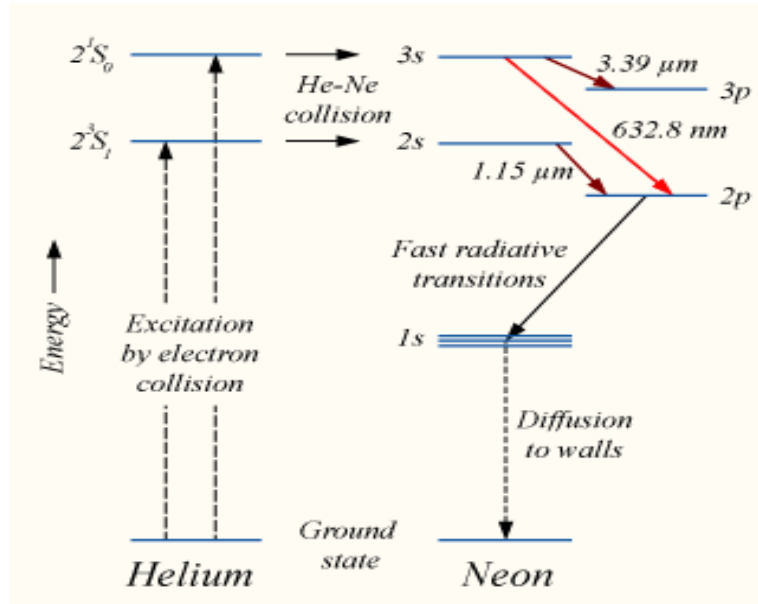


### 3.3.2 He-Ne Laser:

1. It is a 4-level laser system.
2. **Composition:** 10:1 ratio of He:Ne
3. It is a gas laser.
4. Continuous Mode Laser.
5. Electrical Discharge is used for Pumping.
6. **Construction:**



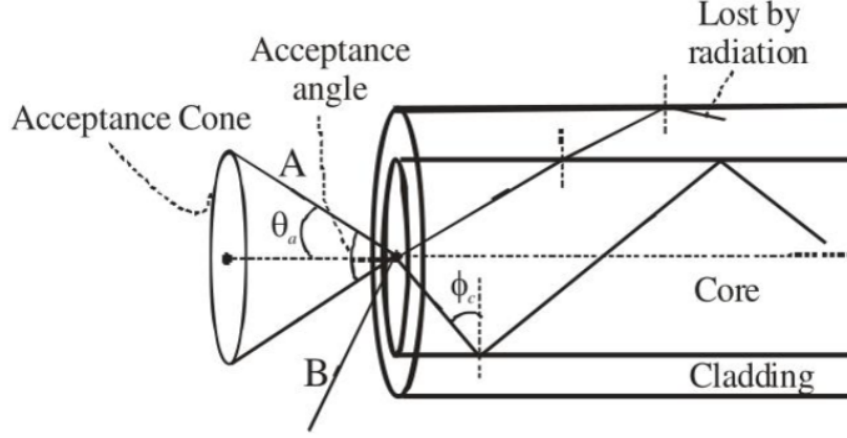
## 7. Energy Diagram:



## 3.4 Optic Fibre:

1. Optical Wave Guide following the principle of "Internal Reflection".
2. It is  $70\mu\text{m}$  or 0.006 inches in diameter.
3. It consists of two coaxial cylinders namely core (inner,  $\text{RI} = n_1$ ) and cladding (outer,  $\text{RI} = n_2$ ). There is also a buffer layer at the outmost.
4. There are two types of Optic Fibres:
  - (i) Step-Index Fiber: Here, Core RI is constant.
  - (ii) Graded-Index Fibre: Here, Core RI is maximum at center, and minimum at sides.

### 3.4.1 Critical and Acceptance Angles:



#### Critical Angle for Total Internal Reflection:

$$\sin \phi_c = \frac{n_2}{n_1} \implies \phi_c = \sin^{-1} \left( \frac{n_2}{n_1} \right)$$

#### Critical Angle for Propagation:

$$\cos \theta_c = \frac{n_2}{n_1} \implies \theta_c = \cos^{-1} \left( \frac{n_2}{n_1} \right)$$

#### Acceptance Angle:

By Snell's Law,

$$n_0 \sin \theta_i = n_1 \sin \theta_c = n_1 \cos \phi_c$$

$$\sin \phi_c = \frac{n_2}{n_1} \implies \cos \phi_c = \frac{\sqrt{n_1^2 - n_2^2}}{n_1}$$

$$\sin \theta_i = n_1 \left( \frac{\sqrt{n_1^2 - n_2^2}}{n_1} \right) \implies \theta_i = \sin^{-1} \left( \sqrt{n_1^2 - n_2^2} \right)$$

#### Cone Angle:

$$\theta_a = 2\theta_i$$

#### Numerical Apperture:

$$NA = \sin \theta_i = \sqrt{n_1^2 - n_2^2}$$

#### Numerical Apperture in terms of Fractional Relative RI ( $\Delta$ ):

$$\Delta = \frac{n_1 - n_2}{n_1} \implies NA = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 + n_2)(n_1 - n_2)}$$

$$\therefore NA = \sqrt{\left( \frac{n_1 + n_2}{2} \right) \left( \frac{n_1 - n_2}{n_1} \right) (2n_1)} \implies NA = n_1 \sqrt{2\Delta}$$

### **3.4.2 Merits of Optic Fibres:**

1. More bandwidth and more speed of transmission.
2. Cost is cheap.
3. Long distance transmission.
4. Small Size, Flexibility, Mechanical Strength, and Light Weight ( $\sim 6\text{kg/km}$ ).
5. Immune to EMI and RFI.
6. No harmful chemical / materials used.
7. No crosstalk, and low loss per unit length ( $\sim 4\text{dB/km}$ ).